

A single-chain derivative of an integrin-activating antibody potentiates organoid growth in Matrigel and collagen hydrogels

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Wim B. M. de Lau^{1,7}, Joost J. A. P. M. Wijnakker^{1,7}, Gijs J. F. van Son², Daniel Krueger¹, Daisong Wang¹, Matthijs S. Abendroth¹, Robin Schreurs¹, Claudia Y. Janda², Fenna L. H. van Rijt^{2,5}, Benjamin J. P. Chalopin², Emma Bokobza², Johan H. van Es¹, Timothy A. Springer³, Arnoud Sonnenberg⁴ & Hans Clevers^{1,6} 

Current methods of culturing human epithelial organoids from adult stem cells may not be compatible with clinical applications as they rely on xenogeneic, chemically undefined or non-standardized components such as the basement membrane extract Matrigel. Matrigel provides a source of extracellular matrix molecules, including laminins and collagen IV, which interact with $\beta 1$ integrins expressed on organoid cells. Here we describe a single-chain (sc) version of antibody TS2/16 that allosterically activates integrin $\beta 1$ function in organoids. The addition of monomeric scTS2/16 to organoid medium results in up to a fivefold increase in the yield of all gastrointestinal organoids grown in Matrigel. Moreover, scTS2/16 supports a six- to sevenfold increase in the yield of these organoids when cultured in collagen I hydrogels, both in 3D and 2D. Collagen I is well defined, available in clinical-grade formulations and, when combined with scTS2/16, may support the clinical application of epithelial organoids derived from gastrointestinal tissues and other epithelial sources.

Organoids are self-organizing, miniature versions of epithelial tissues that can be expanded over long time periods. Since the 2009 report of a 3D technique to culture adult stem cell-derived intestinal organoids¹, protocols have been published for the isolation of organoids representing epithelial tissues derived from all three germ layers¹. However, efforts to apply organoids in areas such as drug screening and clinical transplantation have been hindered by limitations of the culture conditions, which typically include xenogeneic, chemically undefined or non-standardized components.

One essential element of these culture conditions is a growth factor cocktail mimicking the stem cell niche. This typically has to be optimized for each individual tissue, yet always contains three key ingredients: potent activators of the Wnt pathway (for example, Wnt3A or its surrogate, plus R-spondins), a tyrosine kinase receptor activator (for example, EGF) and inhibitors of TGF β and/or BMP (for example, noggin).

A second critical component for proper organoid formation is a cell culture substrate consisting of extracellular matrix (ECM) proteins,

¹Onco Institute, Hubrecht Institute, Royal Netherlands Academy of Arts and Science (KNAW) and UMC, Utrecht, The Netherlands. ²Onco Institute, Princess Maxima Center of Pediatric Oncology, Utrecht, The Netherlands. ³Harvard Medical School, Boston Children's Hospital, Boston, MA, USA.

⁴Division of Cell Biology, Netherlands Cancer Institute, Amsterdam, The Netherlands. ⁵Present address: Synaffix/Lonza Company, Oss, The Netherlands.

⁶Present address: Roche Pharmaceutical Research and Early Development, Basel, Switzerland. ⁷These authors contributed equally: Wim B. M. de Lau, Joost J. A. P. M. Wijnakker. ✉ e-mail: h.clevers@hubrecht.eu

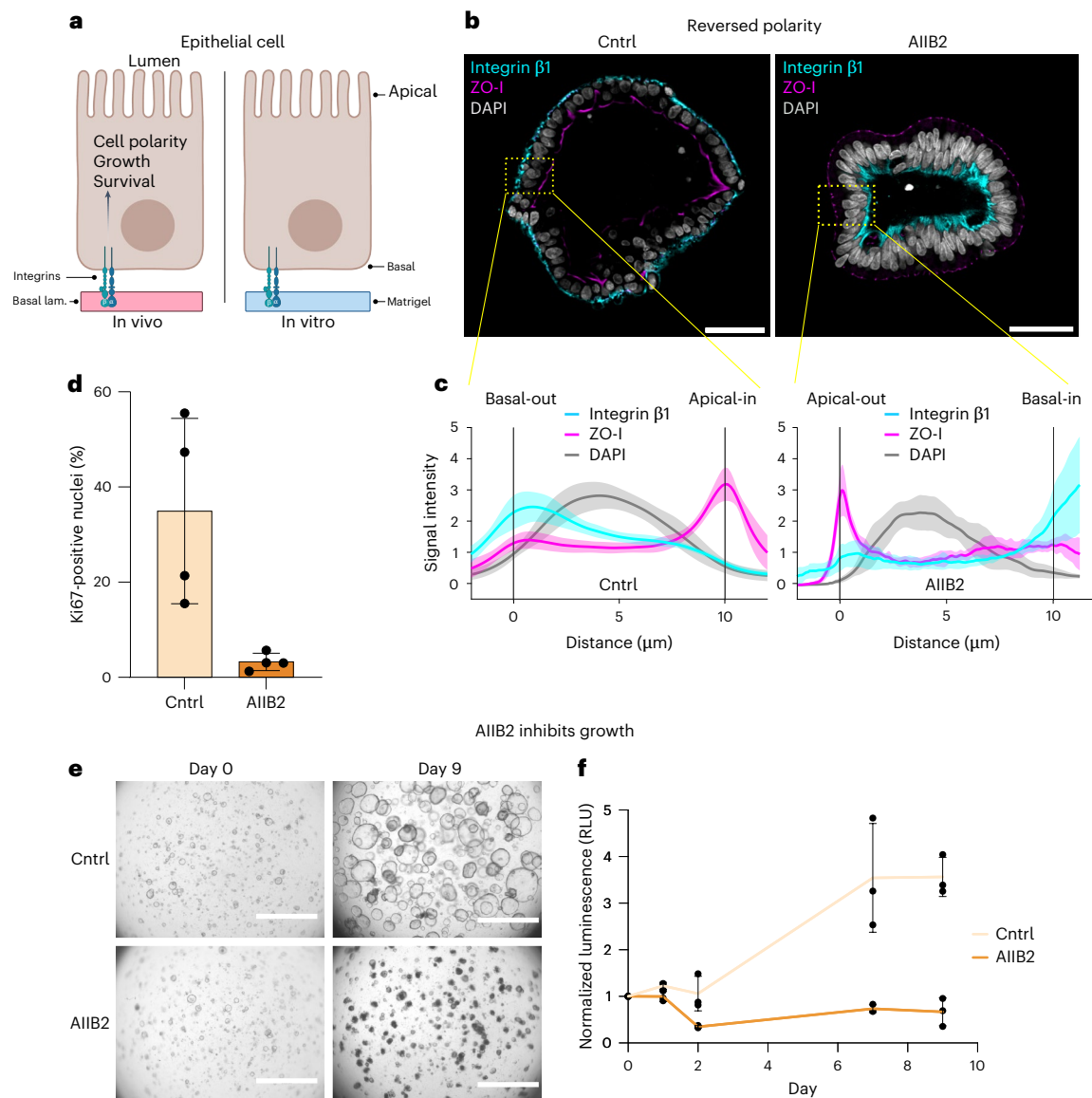


Fig. 1 | Schematic of integrin functionality in epithelial cells in vivo and in vitro. **a**, In vivo, integrin receptors on the basal membrane of epithelial cells contact a mix of ligand proteins of the basal lamina (basal lam.). In vitro, these are replaced by Matrigel or BME proteins. **b**, Organoids were grown in BME in the absence (control, Cntrl) or presence of a strong inhibitory integrin $\beta 1$ Ab (AIB2). Subsequent staining for basal marker integrin $\beta 1$, in both conditions, was performed on organoids with permeabilized cells. The apical marker zonula occludens-1 (ZO-1) and nuclear marker DAPI were used. Yellow marked boxes indicate examples of line scans to quantify the intensity profile of the markers shown in **c**. Scale bars, 50 μm . **c**, Intensity plot of fluorescent apical and basal markers in cross-sections of the membrane. Permeabilized control- and mAb AIB2-treated intestinal organoids as shown in **b** were stained for basal marker integrin $\beta 1$, apical marker ZO-1 and nuclear marker DAPI as a reference. Apical

and basal positions (black vertical lines) were identified based on F-actin staining with phalloidin (not shown), and the mean profile (sharp colored lines) of multiple organoids ($n = 8$ per condition) was calculated (shadowed areas show the s.d.). **d**, Quantification of Ki67-positive nuclei after 2 days of culture of control and mAb AIB2-treated intestinal organoids ($N = 1$ independent experiment), using immunofluorescence staining (one-tailed t -test P value is 8.8×10^{-3} for $n = 4$ biological replicates). **e**, Culture images ($n = 1$ biological replicate) of the survival of intestinal organoids in the presence or absence of AIB2 at days 0 and 9. Scale bars, 2 mm. **f**, For quantification of viable cells ($N = 1$ independent experiment), an ATP-sensitive luminescence assay (CellTiter-GLO) was performed at the indicated days. Mean values for consecutive days represent the mean $\pm$ s.d. ($n = 3$). Data were normalized based on the mean values of day 0.

which in vivo are presented by the basement membrane² (Fig. 1a). This thin layer of a specialized ECM forms the supportive structure for epithelia and includes various laminins (Lns), collagen type IV and, to a lesser degree, arginine–glycine–aspartic acid (RGD) motif-containing molecules such as fibronectin. Lns are heterotrimeric proteins assembled from one of five α chains, one of three β chains and one of three γ chains. In higher organisms, these isoforms show varying degrees of cell and tissue specificity. Collagen IV is a heterotrimeric molecule composed of three α -type chains, which form polygonal networks through several types of intermolecular interactions. Both collagen type IV

and Lns are produced in vivo by epithelial cells and form independent networks that are connected by nidogen and the heparan sulfate proteoglycan perlecan. A highly concentrated blend of these proteins, approximating the composition of the basement membrane, is produced by the mouse Engelbreth–Holm–Swarm tumor after inoculation into the abdomen of mice³. Its major component, embryonic $\alpha 1\beta 1\gamma 1$ Ln (Ln-111), has essential roles in the polarization and differentiation of epithelial cells⁴. The ECM protein collagen type IV is another major component. Engelbreth–Holm–Swarm tumor extracts, collected from mice, are marketed under the names Matrigel and basement membrane

extract (BME). Both are enriched for ECM and, especially Matrigel, have been used extensively for culturing epithelial cells⁴, including 3D structures from mammary epithelium⁴. Matrigel has also enabled the establishment of adult stem cell-based organoid technology⁵. The use of Matrigel is hampered by its cost, batch-to-batch variability, contamination by mouse proteins and ethical considerations on the use of experimental animals. Clinical-grade Matrigel has not been available.

The contact that epithelial cells make in vivo with ECM proteins (or in vitro with Matrigel) is mediated by $\alpha\beta1$ integrins⁶. The specificity of the interaction with individual ECM components is determined by the various α chains, while the resulting intracellular signals are transduced by the $\beta1$ chain. Interactions with Lns proceed via $\alpha3\beta1$ and $\alpha6\beta1$ (ref. 7). The GFOGER motif in collagen allows adhesion in conjunction with $\alpha1\beta1$ and $\alpha2\beta1$ integrins⁶. A third class of ECM proteins, which includes fibronectin, interacts with $\alpha5\beta1$ integrin through the short RGD sequence.

Various studies have described synthetic hydrogels (for example, polyisocyanide (PIC)⁸ or polyethyleneglycol (PEG)⁹, functionalized with integrin-binding motifs taken from ECM proteins. The individual chains of heterotrimeric laminins act cooperatively in binding to integrins with high affinity, and as a consequence, no simple binding motifs have been defined. Therefore, recombinant full-length Ln-111 was coupled to a PIC gel¹⁰. The in vitro synthesis of trimeric laminins in mammalian cells is, however, complex and expensive. By contrast, several synthetic PEG gels have been described in which the GFOGER peptide, alone or in conjunction with the RGD adhesion motif, maintains gastrointestinal organoids, albeit less efficiently than the Matrigel benchmark¹¹. Similar results were obtained with ductal pancreas organoids¹².

To improve the growth of human intestinal organoids in hydrogels, we were inspired by allosteric antibodies activating the integrin $\beta1$ chain operating on leukocytes, first described approximately three decades ago¹³. Like epithelial cells, leukocytes express several $\beta1$ integrins, which mediate interactions with ECM components as well as other cells. Leukocyte adhesion is a highly regulated and essential process in the immune response and inflammation. The human integrin $\beta1$ -specific TS2/16 antibody¹⁴ strongly enhances the adhesion by $\alpha2\beta1$, $\alpha5\beta1$ and $\alpha6\beta1$ integrins with their respective ECM ligands, as well as of $\alpha4\beta1$ to endothelial VCAM protein. A similar TS2/16-induced increase in cell-substrate adhesiveness has been observed for non-hematopoietic cells: for example, neuronal cells in defined stages of chicken embryo development were induced to bind to Ln in vivo in the presence of the $\beta1$ -activating antibody TASC¹⁵.

The non-constitutive nature of integrin interactions implies that ligand binding requires structural changes in the integrins themselves. This was first shown for the $\alpha v\beta3$ and $\alpha IIb\beta3$ integrins¹⁶. Electron microscopy (EM) studies subsequently revealed that the $\beta1$ integrins adopt similar conformational changes¹⁷. Integrins are known to exist in three formational states¹⁶: two low-affinity states, bent-closed and extended-closed, and one high-affinity state characterized by an extended conformation with an open ligand binding domain (headpiece). A variety of $\beta1$ subunit-specific antibodies have been used to induce or stabilize these conformational states and measure their affinity (Supplementary Fig. 1). For example, the anti-integrin $\beta1$ blocking antibody AIIb2 allosterically stabilizes the bent, low-affinity conformation of the $\beta1$ integrin chain¹⁸. By contrast, TS2/16, binds to closed headpieces in the absence of ligand¹⁷ and stabilizes the open, high-affinity headpiece conformation in the presence of ligand^{14,17}. This well-described ability of TS2/16 to enhance leukocyte adhesion inspired us to explore its properties in epithelial organoid cultures. We designed a single-chain variable fragment (scFv) of the original TS2/16 antibody that similarly activates $\beta1$ integrins with the additional advantage of being expressed in bacteria at high yields. The potency of this designed integrin activator in organoid culture is shown across various human epithelial organoids in 3D and 2D in Matrigel and collagen I hydrogels.

Results

Integrin $\beta1$ is indispensable for organoid growth

Adult stem cell-derived intestinal organoids, showing a basal-out phenotype in the presence of external Matrigel, invert their polarity to basal-in when cultured in the absence of Matrigel¹⁹. We tested the role of integrin $\beta1$, dominantly expressed in various organoid models (Extended Data Fig. 1 and Supplementary Fig. 2), on polarity and associated physiology. The rat anti-integrin $\beta1$ antibody AIIb2 has previously been shown to allosterically block the recognition of $\beta1$ integrins for all three major integrin ligand classes (Ln, collagen and RGD-class proteins)¹⁸. This antibody enabled us to comprehensively test the role of the $\beta1$ integrin component in basal-apical polarization of diverse organoid types. Figure 1b–f summarizes the strong effects of adding the AIIb2 antibody to human colon organoids. The ‘basal-out’ phenotype of these organoids, cultured in the presence of BME, inverted within 12 h to a basal-in phenotype following exposure to AIIb2 (Fig. 1b–f). This condition resulted in an immediate abrogation of growth, measured as a strong decrease in the percentage of cells expressing the nuclear proliferation marker Ki67 (Fig. 1d), in visual growth kinetics (Fig. 1e) and in a quantitative cellular ATP assay (Fig. 1f). These observations implied that integrin-mediated signals normally maintain growth and are blocked by AIIb2. Further confirmation for cell cycle exit in response to AIIb2 was seen during a time-course experiment on 2D BME-coated transwells loaded with a human ileal organoid line, stably expressing the fluorescent ubiquitination based cell cycle indicator (FUCCI) reporter (Extended Data Fig. 2a,b). A quantitative PCR analysis, performed on 3D BME-grown duodenal organoids, indicated that AIIb2-arrested cells differentiate into enterocytes and goblet cells (Extended Data Fig. 3).

hIgG and scFv versions of mTS2/16 activate cell adhesiveness through integrin $\beta1$

We next examined the ability of an activating antibody to functionally activate the integrins at the epithelial-organoid cell surface⁶. We constructed a humanized IgG version and single-chain (sc) derivative of the mouse TS2/16 antibody, the latter for ease of production in bacteria (Fig. 2a). We determined the hybridoma-derived sequences of the variable regions of the heavy (V_H) and light (V_L) chains of the mouse Ab TS2/16 (ref. 14) and used human codon-optimized synthetic DNA (Extended Data Fig. 4) to clone these into hulgG1 H- and κ L-chain vectors. The humanized IgG was subsequently purified using standard chromatography purification (Fig. 2b). For the sc version, the combined V regions of the H and L chains were linked in two orientations (Fig. 2a) and connected via a 3×(GGGS) linker. Both protein versions (Extended Data Fig. 5) were cloned into a pET expression vector using *Escherichia coli* codon-optimized DNA. To obtain pure sc versions, a routine protocol for protein extraction, solubilization and refolding was implemented²⁰. On average, we obtained 1 mg pure scTS2/16 per 50 ml of bacterial suspension, sufficient for approximately 7 l of cell culture medium at a final concentration of 5 nM. sodium dodecyl sulfate polyacrylamide gel (SDS-PAGE) analysis of the purified scFv version of TS2/16 (V_H – V_L) is shown in Fig. 2b. We also developed a protocol to produce scTS2/16 in the human cell line HEK293 (Extended Data Fig. 6a–c). The scTS2/16 (V_H – V_L) bound the ectodomain of the $\beta1$ subunit in solution, analogous to its parental IgG version (Extended Data Fig. 7a,b). Functionally, both hulgG1 and scFv versions (V_H – V_L) induced adhesion at low nM concentrations of K562 (erythroleukemia) cells to fibronectin, a standard assay for integrin-mediated adhesion (Fig. 2c,d). We decided to focus on the scFv (V_H – V_L) version of TS2/16 (scTS2/16) for subsequent studies.

scFv-TS2/16 enhances growth of intestinal organoids in BME

One of the main components of BME is the protein Ln-111. Four different integrins, $\alpha3\beta1$, $\alpha6\beta1$, $\alpha7\beta1$ and $\alpha6\beta4$, differentially bind to members of the Ln family of proteins²¹. Integrin $\alpha6\beta1$ (ref. 22) binds to Ln-111, the

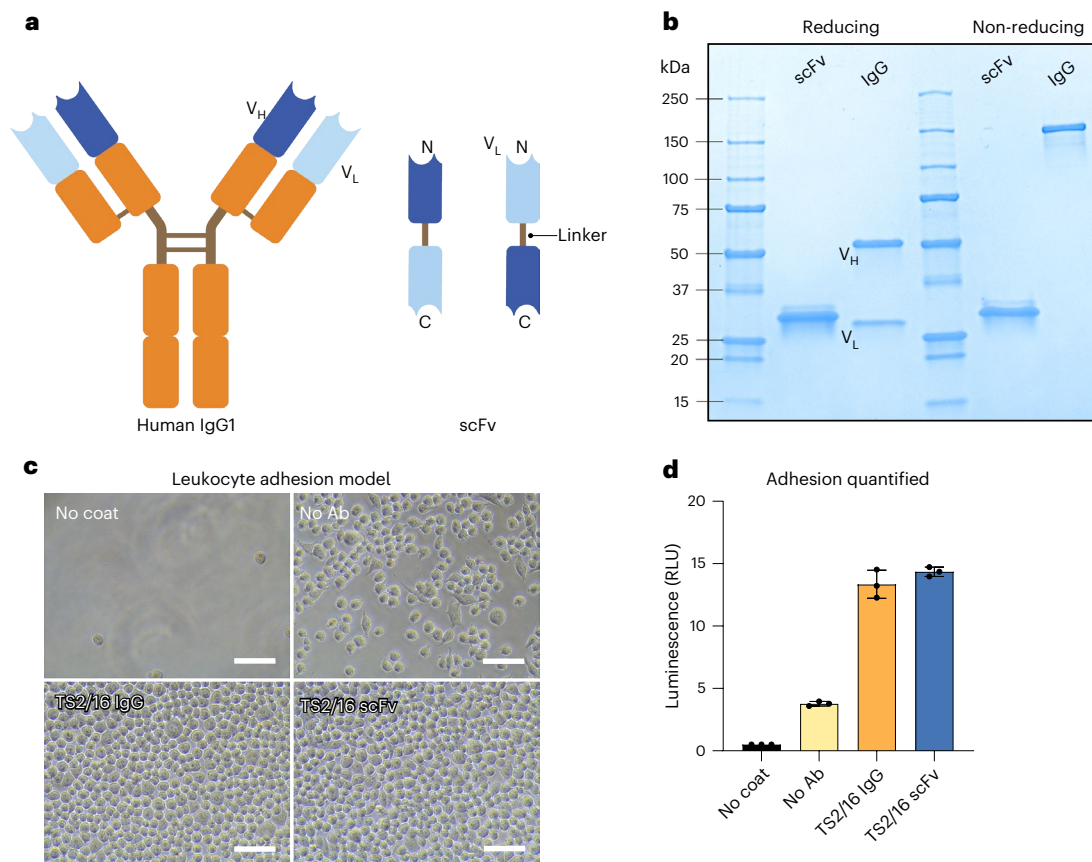


Fig. 2 | Classical IgG and scFv versions of the activating $\beta 1$ integrin antibody TS2/16. **a**, Integrin recognition-providing V_H and V_L chains of the mouse hybridoma TS2/16 were cloned into a human IgG1 backbone or combined in a single-chain antibody version. Both scFv orientations (V_H-V_L and V_L-V_H) are shown with the N-terminal and C-terminal. The concept of the procedure is schematically displayed. **b**, SDS-PAGE analysis of purified TS2/16 IgG (± 180 kDa) and scFv (± 26 kDa) (shown is the V_H-V_L version) under reducing and non-reducing conditions ($N = 2$). Proteins were visualized by Coomassie Blue staining. The positions of separate H and L chains of IgG under reducing conditions are indicated. The molecular weights of markers in kilodaltons (kDa) are indicated on

the left margin. **c**, Bright-field images of the K562 leukocyte cell-based adhesion reporter assay. K562 cells were allowed to adhere to plastic-coated fibronectin for 1 h at 37 °C in the absence or presence of 5 nM of IgG- or scFv-type TS2/16 antibody. Cells, remaining after three buffer washes under the conditions indicated, are shown (triplicate values). $N = >3$ for number of experiments. Scale bars, 100 μm. **d**, A luminescence cell viability assay (CellTiter-Glo) was used to quantify the adhesion of K562 cells ($N = 3$ for number of experiments) to the fibronectin-coated plate under conditions shown in **c**. RLU values are in millions; means of triplicate values ($n = 3$) and s.d. are indicated (one-tailed t -test $P = \pm 6.6 \times 10^{-5}$ for IgG TS2/16 versus no Ab, and 7.7×10^{-7} for ctr versus scTS2/16).

embryonic Ln present in BME. The organoid-expressed $\alpha 3\beta 1$ and $\alpha 6\beta 4$ integrins bind to Ln-332 and Ln-511/521 (Extended Data Fig. 8), but these are absent in BME. The specific role of $\alpha 6\beta 1$ integrin in Ln-111 binding was confirmed by exogenous expression of integrin $\alpha 6\beta 1$ in K562 cells. Ln-111 recognition acquired following integrin- $\alpha 6$ transfection was evident, but certainly not maximal. It could be strongly enhanced by adding the activating TS2/16 antibody²³.

We tested whether this phenomenon would translate to potentiated adhesion of colon organoid cells to coated BME. We used a series of concentrations of scTS2/16 in the low nM range. The adhesion in the absence of scTS2/16 (11×10^5 relative light units (RLU)) reached its maximum (22×10^5 RLU) at 2 nM (Fig. 3a). The AIB2 blocking antibody inhibited the spontaneous adhesion of organoid cells, as a validation of this adhesion assay. As BME contains both Ln-111 and collagen IV, we tested the contributions of these components independently in organoid adhesion. Again, we detected maximum enhancement effects of scTS2/16 at 2 nM for the individual components. The inhibitory AIB2 antibody completely inhibited the adhesion of organoids to both coats (Supplementary Fig. 3). To assess whether the integrin $\alpha 6$ subunit was also crucial in mediating growth of colon organoids in BME, we added the inhibitory integrin $\alpha 6$ -specific GoH3 antibody²⁴ while regularly splitting colon organoid cultures. The results shown in Fig. 3b indicated a strong GoH3-induced reduction of growth. We then tested whether

activation of the $\beta 1$ integrins could enhance the growth of colonic organoids. Addition of scTS2/16 to standard 3D colon organoid culture in BME at 5 nM led to a 3-fold increase in colon organoid growth (0.9×10^6 to 2.8×10^6 RLU). This was observed after 10 days of culture by assessment of total genomic DNA, ATP-dependent luminescent light units (Fig. 3c) and bright-field images of the organoid cultures (Fig. 3d). An essential component of integrin adhesion is the presence of cations of manganese (Mn^{2+})^{21,25}. We supplemented 1 nM $MnCl_2$ -containing, complete organoid medium with 1 mM $MnCl_2$. The latter serves as the benchmark for maximal integrin adhesion. We then directly compared the Mn^{2+} effects on adhesion and proliferation to 5 nM scTS2/16. While both induced an increase in adhesion, only scTS2/16 potentiated growth (Supplementary Fig. 4b). The proliferation-promoting effects of scTS2/16 are reflected in the increase of Ki67-positive nuclei from 20% in the absence of the scFv to 36% in the presence of the scFv ($P = 4.1 \times 10^{-4}$) (Supplementary Fig. 5).

Single-cell cloning from primary tissue or from organoids is generally highly inefficient^{5,26}. We tested the effect of scTS2/16 on the plating efficiency of such single cells and noted that the clonal outgrowth in the presence of scFv TS2/16 (at 5 nM) improved from 30×10^3 RLU in the absence of scTS2 to 330×10^3 RLU in the presence of scTS2 (Fig. 3e,f). By contrast, the outgrowth of organoids, even in the presence of scTS2/16, was completely blocked by adding the inhibitory integrin $\alpha 6$ -specific

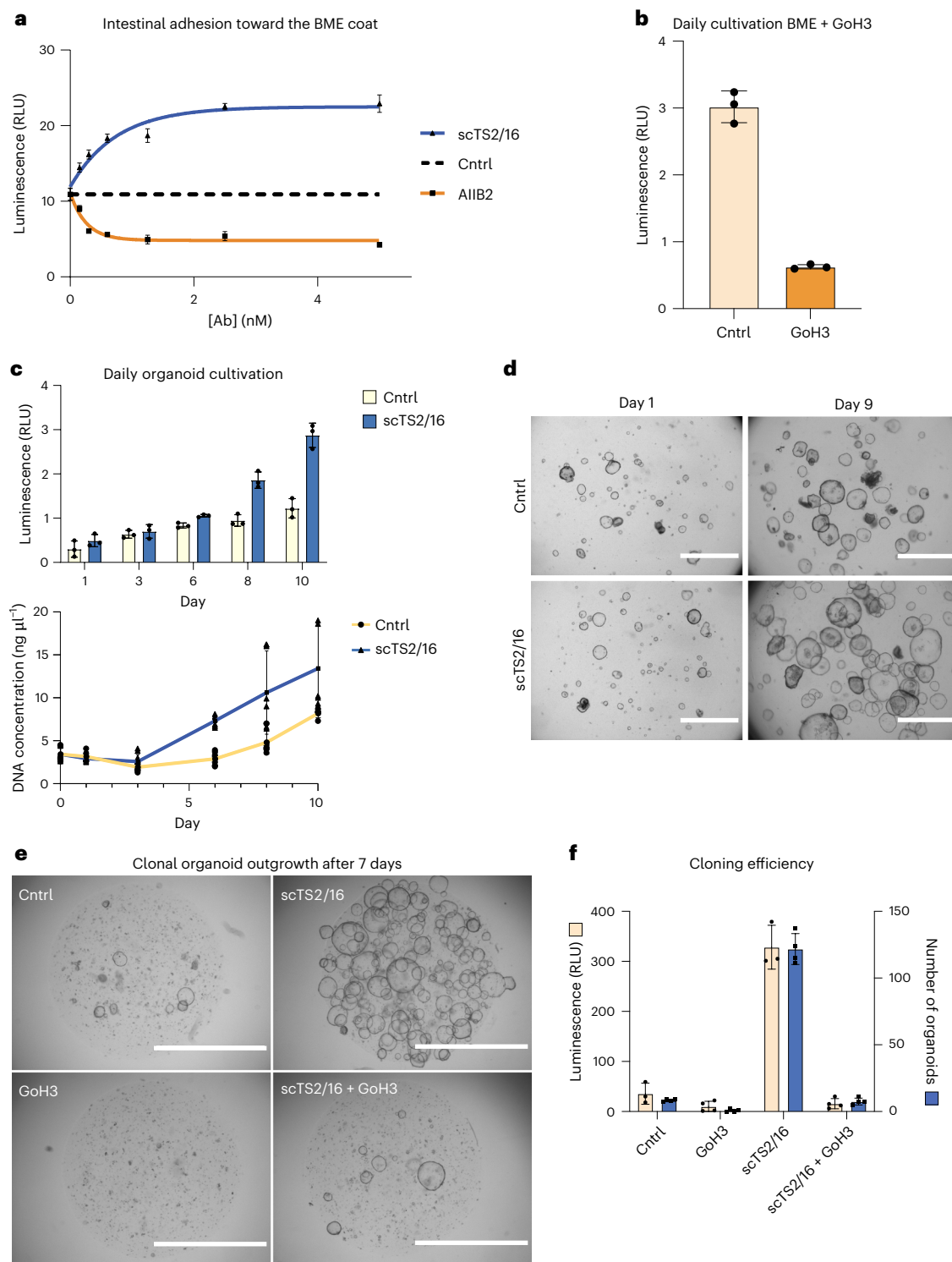


Fig. 3 | scFv TS2/16 enhances growth of intestinal organoids in BME. a, Quantification of colon organoid-derived single cells adhering to coated BME expressed as $\text{RLU} \times 10^5$ from a cellular ATP-driven luminescence assay (CellTiter-Glo). Test conditions include the absence of antibody and nM concentrations of the activating scTS2/16 and the inhibitory mAb AIIB2. The graph represents trend lines based on mean $\pm$ s.d. ($n = 3$ biological replicates) for each condition (independent experiments, $N \geq 3$). **b**, Routine cultivation of identically processed fragmented intestinal organoids in the absence or presence of integrin subunit $\alpha 6$ blocking mAb GoH3. Quantification of ATP-driven luminescence assay ($\text{RLU} \times 10^5$) after 7 days of culturing is presented as mean $\pm$ s.d. ($n = 4$ biological replicates) (one-tailed t -test $P = 3.4 \times 10^{-5}$ for cntrl versus GoH3). **c**, Routine cultivation of identically processed fragmented intestinal organoids in the absence or presence of scTS2/16. Mean $\pm$ s.d. ($n = 4$ biological replicates) of

total genomic DNA (y-axis) and relative luminescence ($\times 10^6$) in the cell viability assay at indicated days are depicted. **d**, Bright-field culture images of intestinal organoids from the experiment shown in **c**, as present on days 1 and 9. Scale bars, 2 mm. **e**, Culture images of a cloning assay, starting with a very dilute single-cell suspension. Control, GoH3 and scTS2/16-supplemented medium are indicated. Scale bars, 2 mm. **f**, Quantification of the number of organoids (right y-axis) scored after 7 days from organoids as presented in **e**; the left y-axis shows cell viability expressed as $\text{RLU} \times 10^3$ from a cellular ATP-driven luminescence assay (CellTiter-GLO). Mean ($n = 3$ biological replicates) $\pm$ s.d. is indicated. P values one-tailed t -tests comparing the luminescence of the control versus that of GoH3, scTS2/16 or both are 8.7×10^{-2} , 2.0×10^{-4} and 1.3×10^{-1} , respectively. For number of cells, these values are in succession 1.6×10^{-5} , 5.7×10^{-7} and 2.9×10^{-1} . Number of experiments $n = 1$.

GoH3, emphasizing the importance of the integrin $\alpha 6$ component. The positive effects of integrin $\beta 1$ -stimulatory antibodies on the outgrowth of organoids from single cells were much less strong for the two other $\beta 1$ -chain conformation-stabilizing antibodies, 9EG7 (ref. 27) and 12G10 (ref. 28) (number of clones 15, 35, 20 and 100 at 5 nM for the control, 12G10, 9EG7 and TS2/16, respectively (Supplementary Fig. 6). The TS2/16-stabilized high-affinity conformation of $\beta 1$ integrins on leukocytes enhances intercellular (VCAM-1)²⁹ and cell-to-ECM adhesion^{17,30}. Our data show that an scFv version of the antibody also improves the establishment and expansion of epithelial organoids by the enhancement of $\alpha 6\beta 1$ -mediated interaction with Ln-111/collagen IV as presented in the BME.

BME-established organoids from different regions of the gastrointestinal tract also rely on integrin $\beta 1$

We then tested whether the observed integrin $\beta 1$ -regulated proliferation of intestinal organoids in BME also held for organoids derived from other regions of the human gastrointestinal tract: duodenum ($N = 2$), J2 jejunum ($N = 2$) and N39 ileum ($N = 2$), stomach organoids³¹ ($N = 1$) and ductal-type organoids from the pancreas³² ($N = 2$) and liver³³ ($N = 2$). The pertinent organoids were grown in BME supplemented with control medium or with 5 nM of either the inhibitory mAb AIIB2 or the stimulating scTS2/16 antibody. AIIB2-induced inhibition and scTS2/16 enhancement of growth (compared with the medium without antibody) were observed in all cases (fold increases in examples shown in Fig. 4 were 5-fold (stomach), 2.2-fold (duodenum), 1.6-fold (liver), 8.1-fold (pancreas), 1.5-fold (jejunum) and 1.4-fold (ileum)) (Fig. 4). To exclude major morphological changes resulting from the use of scFv, high-magnification confocal images of 3D BME-scTS2/16-grown jejunal organoids are presented in Supplementary Fig. 7. To assess whether scTS2/16-stimulated growth caused shifts in the transcriptome of the cells, bulk sequence experiments were performed on control and Ab-stimulated 3D BME cultures of P26N colon, Duo-72 duodenum and J2 jejunum organoids (Extended Data Fig. 9). The statistical analysis revealed no significant changes. We concluded that growth of gastrointestinal tract organoids depends on $\beta 1$ -class integrins and is strongly enhanced by the addition of scTS2/16 to their medium.

scTS2/16 antibody potentiates growth of established and primary isolated colon organoids in collagen I hydrogel

Based on the scFv-generated improvement of gastrointestinal organoids in BME, we tested the hypothesis that organoid growth in an alternative hydrogel would also benefit from the addition of scTS2/16. Collagen recognition by integrins was shown to be highly sensitive to TS2/16 (ref. 34) and has been explored extensively for in vivo and in vitro applications^{35–37}. In a first attempt to enhance growth by scTS2/16 in a collagen type I hydrogel, we focused on colon organoids. A commercially available preformulated isotonic form of collagen I with neutral pH forms a 3D hydrogel by simply warming to 37 °C (PureCol EZ Gel). In total, 4 α -integrin subunits ($\alpha 1$, $\alpha 2$, $\alpha 10$ and $\alpha 11$) harbor a collagen-binding I-domain insert and share the $\beta 1$ subunit. Of these, we observed the expression of $\alpha 2\beta 1$ in BME-grown colon organoids, both at the RNA level (Extended Data Fig. 1) and by immunofluorescence staining (Fig. 5a). In accordance with these findings, the scTS2/16 amplified adhesion of colon organoid cells toward PureCol EZ Gel (uninduced 7.5×10^5 to 22.5×10^5 when scTS2/16-induced) (Fig. 5b). By contrast, the inhibitory AIIB2 antibody blocked the uninduced and induced components of this adhesion (Fig. 5b). P26N colon organoids showed a ± 3 -fold increase of their growth in collagen (Supplementary Fig. 8b). Also in these colon organoids, no major morphological changes were observed in confocal images (Supplementary Fig. 8c).

We then created null alleles of the gene for the integrin $\alpha 2$ subunit (*ITGA2*) by CRISPR to confirm the relevance of the collagen-recognizing ($\alpha 2\beta 1$) integrin for colon organoid growth in collagen hydrogels (Fig. 5c). As predicted, *ITGA2* gene mutation blocked organoid growth

in collagen I, even in the presence of scTS2/16. While BME enabled isolation of the *ITGA2*^{−/−} cells, growth was slightly reduced compared with the parental organoids, indicating that BME growth of organoids is at least in part controlled by collagen IV (Fig. 5d).

We then tested fresh colon biopsies for organoid outgrowth in collagen I hydrogel, both in the presence and absence of scTS2/16. Colon tissue from donors ($N = 3$) was enzymatically processed and epithelial cell suspensions were embedded in collagen I. The addition of 5 nM of scTS2/16-containing medium yielded substantial more organoids (average of 25 clones; $n = 4$) compared with standard colon medium (average of 6 clones; $n = 4$) (Fig. 5e). Both BME-established organoid lines, as well as those from primary biopsies, showed difficulty expanding in collagen hydrogel after repeated splitting. A ‘niche-inspired’ medium was recently designed for intestinal organoids³⁸. We tested the contribution of the growth factors IGF1 and FGF2 (taken from this protocol) for application in collagen hydrogel (of note, the relevant receptors are expressed at the mRNA level). Each of these contributed to improved growth. In addition, CHIR and forskolin further enhanced the culture. We compared growth (percentage Ki67) and apical–basal polarity (ezrin and integrin $\beta 1$, respectively) of an ileum organoid line in three hydrogel conditions: BME, collagen and collagen^{scTS2/16}. At passage 1, organoids growing in the absence of scTS2/16 lost their apical–basal polarity as well as the proliferation marker Ki67 (Supplementary Fig. 9). We then tested the same culture conditions growing an ileum organoid line harboring fluorescent reporters for all three secretory cell types, enabling monitoring of ongoing differentiation. These N39 triple-reporter organoids could be passaged at least six times in collagen (Supplementary Fig. 10d), but appeared fully dependent on it for the addition of scTS2/16. Cells from passages P0 and P2 showed differentiation toward all three secretory cell types when subjected to differentiation medium. Supplementary Fig. 10a presents fluorescent confocal images of the reporters while Supplementary Fig. 10b provides a quantitative analysis of these reporters based on flow cytometry.

When ITGA2-expressing human liver bile duct organoids (Extended Data Fig. 1) were stimulated by scTS2/16, increased adhesion toward collagen I ensued. Guided by these results, we tested the outgrowth of cholangiocyte organoids in collagen I hydrogels, directly from primary liver biopsies. This mirrored the adhesion results. There were consistently more organoids in the scTS2/16 medium compared with the control medium from 3 to 13 (ctr versus scFv) and from 13 to 22 (ctr versus scFv), respectively ($n = 2$) (Supplementary Fig. 11). These proved to be epCAM^{high}, as expected for this cell type. We subsequently quantified the scTS2/16 effects on the newly derived collagen-based cholangiocyte organoids (Fig. 5f). A substantial twofold increase in number of organoids (ctr 15 to 35) was seen for ctr and scTS2/16, respectively.

Based on the principles established in the 3D cultures, we tested whether collagen-based culturing of organoids in 2D would similarly be enhanced by scTS2/16. Such cultures have been described previously, in which intestinal cells were cultured on a monolayer of myofibroblasts³⁹. Other protocols allow short-term human organoid growth⁴⁰ or apply a more complex collagen sandwich technique in combination with the use of small molecules (CHIR and valproic acid) to maintain growth⁴¹.

Here we tested scTS2/16-supported growth of P26N colon, Duo-72 duodenal and J2-jejunal organoids in transwells. These were coated with the same collagen used in 3D. Growth was compared with that in BME-coated inserts under otherwise identical conditions. Results are shown in Supplementary Fig. 12 (P26N), Supplementary Fig. 13 (duodenum) and Supplementary Fig. 14 (jejunum). scTS2/16-supplemented P26N colon cultures produced 2.8-fold to 4.5-fold more tissue, on BME and collagen, respectively. For the Duo-72 duodenum cultures, improvements in response to scTS2/16 were measured both on BME (amplification ± 2.4 -fold) and collagen (amplification ± 2.4 -fold). For the jejunum line (Supplementary Fig. 14), these amplification factors were ± 7.5 (BME) and ± 4.75 (PureCol EZ Gel). Similarly to the 3D cultures,

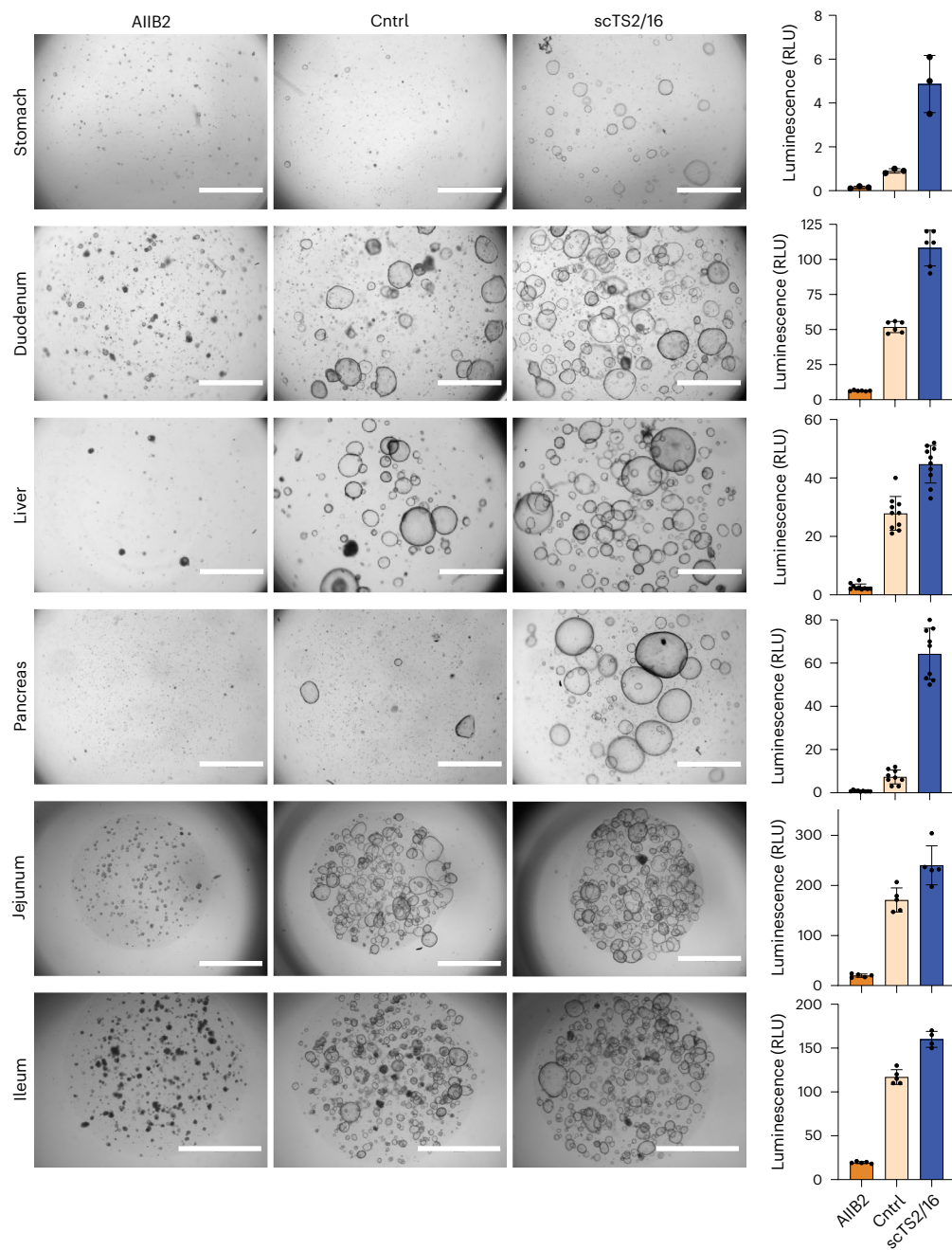


Fig. 4 | BME growth of gastrointestinal tract organoids is mediated by $\beta 1$ integrins. Single-cell initiated cultures of stomach ($N = 1$), duodenum ($N = 2$) and ductal organoids from the liver ($N = 2$) and pancreas ($N = 2$), jejunum ($N = 2$) and ileum ($N = 2$) were grown in the presence of either 5 nM inhibitory mAb AIB2 or activating scTS2/16. The organoid and medium types are indicated at the left and top of the figure, respectively. Scale bars, 2 mm. For quantification of the number of viable cells, an ATP-sensitive luminescence assay (CellTiter-GLO) was performed on the same day the images were taken. The mean RLU ($\times 10^5$) $\pm$ s.d. is indicated. One-tailed t -test P values for the control versus AIB2 and scTS2/16

are 1.5×10^{-4} and 3.1×10^{-3} for the stomach (number of experiments, $N = 1$; $n = 3$ biological replicates). Similarly for duodenum, 4.0×10^{-11} and 5.8×10^{-7} (number of experiments, $N = 2$; $n = 6$ biological replicates); liver, 3.1×10^{-11} and 4.2×10^{-6} (number of experiments, $N = 4$; $n = 10$ biological replicates); pancreas, 9.0×10^{-6} and 1.1×10^{-10} (number of experiments, $N = 5$; $n = 9$ biological replicates); jejunum, 6.0×10^{-8} and 2.8×10^{-3} (number of experiments, $N = 2$; $n = 5$ biological replicates); and finally, ileum 3.0×10^{-4} and 7.67×10^{-5} (number of experiments, $N = 2$; $n = 5$ biological replicates).

we tested the stability of the transcriptomes of these 2D cultures in the presence of scTS2/16. Bulk RNA sequencing confirmed the stability of these in vitro intestinal cultures (Extended Data Fig. 10).

Given these positive short-term 2D results, we then tested whether these effects would be maintained in long-term culture. We first cultured Duo-72 duodenum organoids in collagen-coated transwell inserts in the presence and absence of scTS2/16 (Fig. 6a). We passaged both types of cultures four times, restarting with a fixed number of single

cells at each round. The experiment was quantified by scoring the number of cells in inserts at the end of each passage. By extrapolating these values over four passages, we found that approximately 15 times more cells were produced in the presence of scTS2/16. For the second experiment (Fig. 6b), we cultured J2 jejunum organoids in collagen-coated inserts for six passages in the presence of scTS2/16. We then tested the competency of these cells to differentiate to the major specialized intestinal cell types. Cells, sequentially for 2 weeks in

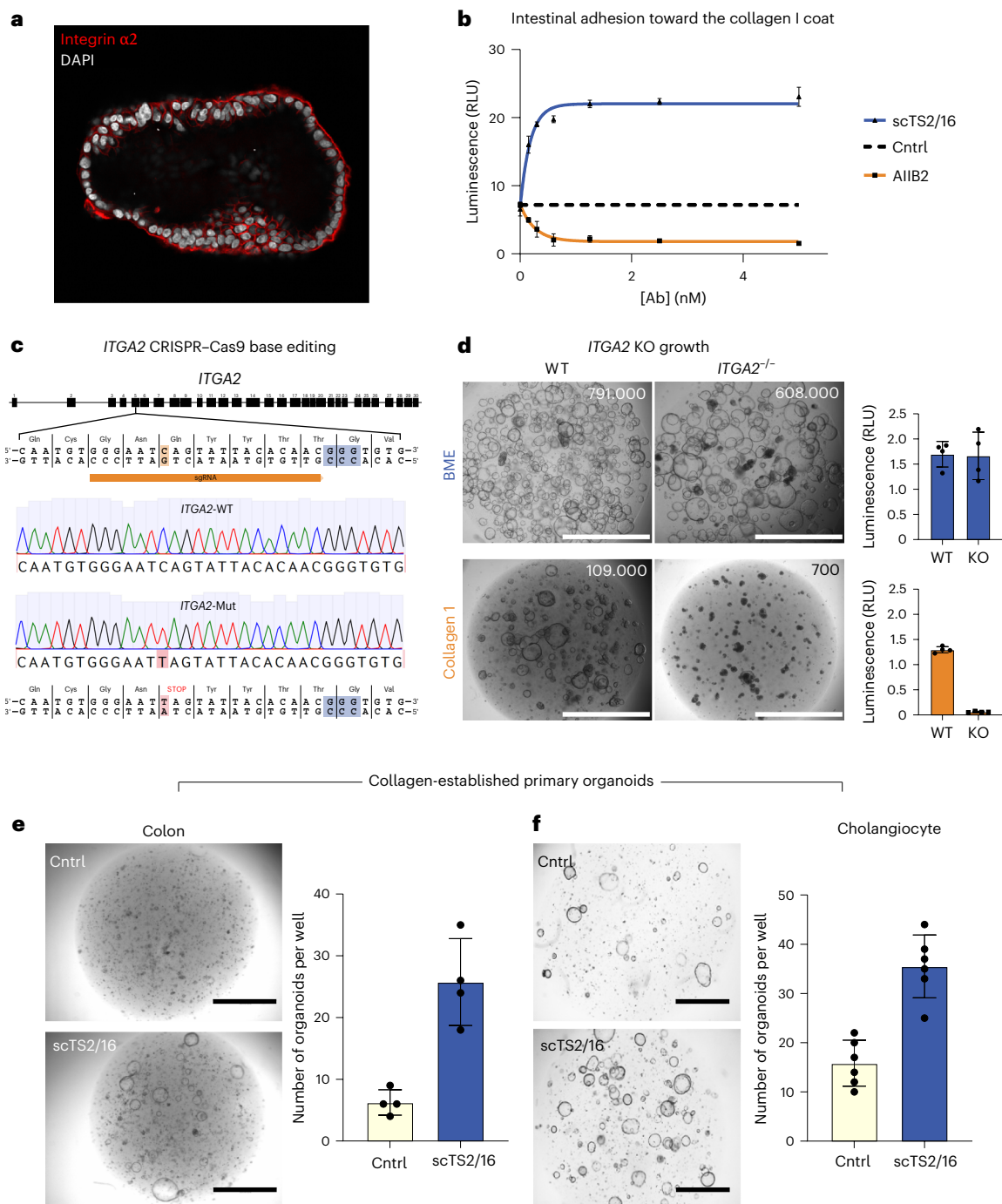


Fig. 5 | scTS2/16 improves integrin $\alpha 2\beta 1$ -mediated growth of primary organoids in PureCol EZ Gel. **a**, BME-grown intestinal organoids stained for integrin subunit $\alpha 2$ and nuclear marker DAPI ($N = 1$). Scale bar, 50 μm . **b**, Quantification of intestinal organoid-derived single cells adhering to coated collagen I, using a cellular ATP-driven luminescence assay (CellTiter-Glo). Test conditions include absence of antibody and a series of nM concentrations of the activating TS2/16 single-chain Fv and the inhibitory mAb AIIIB2. The graph represents trend lines based on mean $\pm$ s.d. ($\text{RLU} \times 10^5$) ($n = 3$ biological replicates) for each condition. **c**, Sanger sequence validation of *ITGA2* gene KO using CRISPR–Cas9 base editing (base C>T editing). Editing transforms a Gln amino acid into a stop codon. *ITGA2* exons are illustrated to identify the location of the designed sgRNA (orange). The targeted base is highlighted in orange, and the PAM sequence is identified in blue. Single colonies were grown in the presence of scTS2/16 and picked for sequencing. The image represents the relevant DNA sequence of 1 of 3 verified clones. **d**, Parental and *ITGA2*-KO

organoids were grown in parallel in BME and PureCol EZ Gel, in the presence of scTS2/16 ($N = 3$ for number of experiments). Growth was quantified by CellTiterGlo. Mean $\pm$ s.d. of the relative light units ($\text{RLU} \times 10^3$) of biological replicates ($n = 4$). In parallel wells ($n = 1$ biological replicate), tissue yield was measured by counting cells (numbers shown in the images). **e, f**, Primary biopsies ($N = 3$ experiments) of colon (**e**) and cholangiocyte ($N = 2$ experiments) (**f**) tissue were processed and embedded in collagen I in the presence or absence of scTS2/16. For the colon, images and quantification were done after 10 days of culture ($n = 4$ biological replicates). P value for control versus scTS2/16 = $\pm 3 \times 10^{-3}$. Scale bars, 2 mm. For primary cholangiocytes, collagen-derived organoids (Supplementary Fig. 11) were cultured for 3 weeks. Subsequently, scTS2/16 sensitivity was measured ($N = 1$ for number of experiments, with 6 biological replicates) by counting the number of cells (P value one-tailed t -test for scFv effect 5.8×10^{-5}).

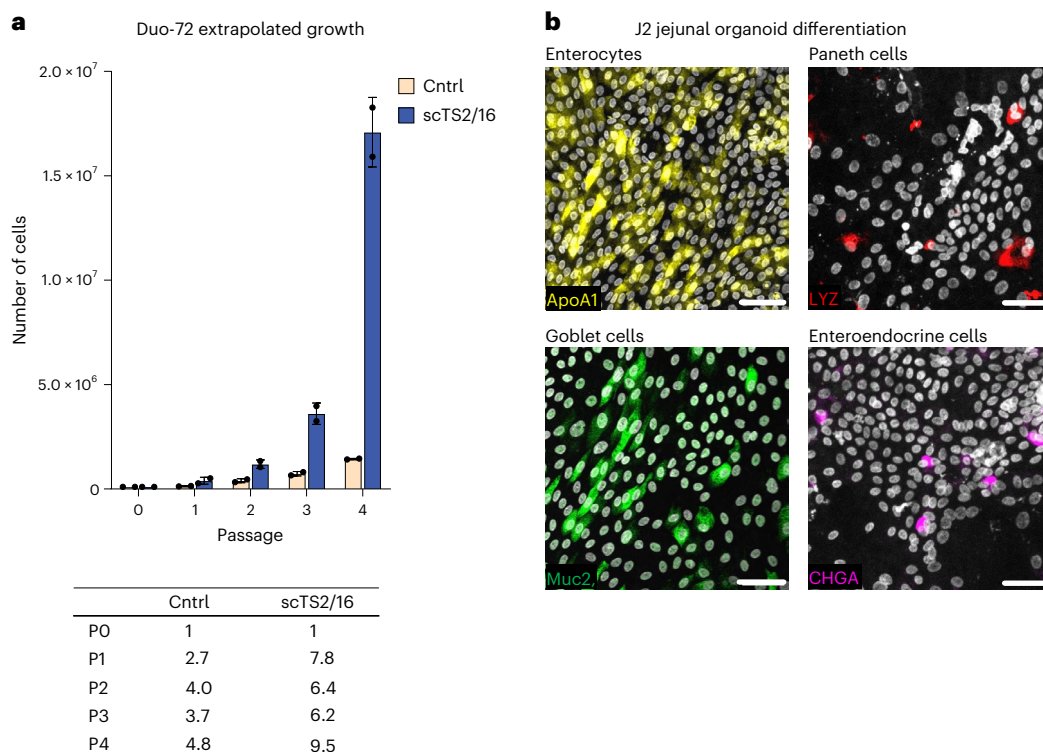


Fig. 6 | Long-term 2D growth and differentiation of intestinal organoids.

a. Jejunal organoid cultures were started with 5×10^4 cells ($n = 2$) on PureCol EZ Gel-coated inserts of transwells. Intestinal expansion medium, $\pm$ scTS2/16, was present in the upper and lower compartments. Passaging of control and scFv cultures took place after approaching full confluency in the latter. In both control and scTS2/16 conditions, cells were counted and gain factors calculated (table at the bottom). Freshly coated inserts were reloaded with 5×10^4 cells. Cumulative growth as depicted in the graph was calculated by multiplying the measured

gain factors in successive passages. **b.** Duodenal organoids were scTS2/16 grown in collagen I-coated transwells. After 6 passages, the expansion medium was exchanged with 'patterning' medium for 14 days, and subsequently for 14 days with 'maturation medium'. Both scTS2/16-supplemented differentiation media were applied only to the bottom compartment (liquid–air culture). The panels show confocal images of ApoA1 staining in enterocytes, lysozyme (LYZ) in Paneth cells, mucin 2 (Muc2) in goblet cells and chromogranin A (CHGA) in enteroendocrine cells.

patterning⁴² and 2 weeks in IL22⁺ maturation⁴² media (both scTS2/16⁺), had retained the capacity to differentiate toward enterocytes, goblet cells, Paneth cells and enteroendocrine cells (Fig. 6b). The results were found to be indistinguishable from the same cells grown under the same conditions but on a BME coat.

Discussion

Here we exploit the functional importance of integrin $\beta 1$ in epithelial organoid cultures derived from the human small intestine, colon, stomach, pancreas and liver. We describe an sc integrin-activating antibody, scTS2/16, that is easily produced and strongly enhances organoid outgrowth. We observe potent growth-blocking effects of the inhibitory integrin $\beta 1$ subunit-specific antibody AIIB2, while the activating integrin $\beta 1$ subunit antibody TS2/16 enhances organoid growth. The agonistic scTS2/16 binds the human $\beta 1$ subunit with high affinity. Moreover, scTS2/16 is highly potent in supporting the establishment and growth of organoids in the standard BME hydrogel, simply by addition to the culture medium as a soluble monomer. We observe a crucial role for the integrin $\alpha 6$ component in BME-dependent growth. We currently believe that organoid growth in BME results from the parallel interaction of $\alpha 2\beta 1$ with collagen IV and $\alpha 6\beta 1$ with Ln-111, both natural ECM components present in BME. We also find that scTS2/16 supports the growth of organoids in hydrogels of purified collagen I. CRISPR-mediated *ITGA2*-gene knockout showed the $\alpha 2\beta 1$ integrin to be responsible for this effect. Our 2D protocol reveals that scTS2/16 enables potent, long-term culturing of intestinal tissue on collagen coats. The tissue generated under these conditions remains competent to differentiate into secretory Paneth cells and cells participating in the absorption of nutrients

(enterocytes) and the production of mucus (goblet cells) and hormones (enteroendocrine cells).

The TS2/16 antibody reinforces adhesion and growth in organoids most probably by providing a counterforce to contractility of the cytoskeleton. A strong indication of this is the inhibition of collagen contraction by TS2/16 (ref. 43), identical to the effect of Rho-associated coiled-coil containing protein kinase (ROCK) inhibitors⁴⁴. Involvement of ROCK inhibition in growth was shown in primary keratinocytes, which are conditionally transformed following prolonged exposure to small-molecule inhibitors of the ROCK kinase⁴⁵. Functional evidence for the initiating role of integrin–ligand signaling in these effects was nicely shown by deleting *ITGB1* in preimplantation epiblasts⁴⁶. The versatility of scTS2/16 arises most probably from the fact that the integrin α chains of the ECM proteins Ln and collagen I or collagen IV share the $\beta 1$ signal transduction chain.

Several earlier studies describe well-defined hydrogels for gastrointestinal organoids. Generally, these studies have used organoids established in Matrigel as starting material. The first two involved synthetic PEG gel^{11,12}, functionalized with ligand mimetic peptides PHSRN-RGD (Fn) and GFOGER (Col), combined with a mix of ECM-sequestering peptides. In one of the studies¹¹, growth of human small intestinal organoids was successful, although the gel did not match the Matrigel benchmark of the organoid emergence rate and overall proliferation. The collagen mimetic GFOGER peptide clearly dominated in driving proliferation over the PHSRN-RGD peptides. This is in line with our collagen results and with fibrinogen hydrogel findings by others⁴⁷. The second study describes growth of normal mouse ductal pancreas organoids, and mouse and human adenocarcinoma organoids¹². In these three organoid types, growth was again

largely dependent on the GFOGER peptides. Notably, growth was reduced in the presence of antagonists of laminin- $\alpha 3\beta 1$ or $\alpha 6\beta 1$ interactions in adenocarcinoma-type organoids. It is unclear whether this Ln-driven growth is a characteristic of the adenocarcinoma status, is specific for mouse species or is a result of the PEG-bound ECM-binding peptides retaining cell-secreted laminins. It thus remains unclear whether normal human pancreas organoids would also grow in this gel. An obvious possibility is to supplement such systems with scTS2/16, thereby improving integrin interaction with both ligand mimetics and cell-secreted ECM proteins. A PIC hydrogel, decorated with high concentrations of Matrigel isolated laminin-entactin, or recombinantly produced laminin-111, successfully maintained human liver organoid cultures¹⁰. It appears unlikely that this PIC-recombinant Ln-111 hydrogel would provide an affordable hydrogel. One previous study has described that human colon organoids can be grown in collagen I directly from primary tissue³⁹. We predict that this approach will benefit from the addition of scTS2/16.

The *Yersinia* protein invasin is known to activate $\alpha 3\beta 1$, $\alpha 4\beta 1$, $\alpha 5\beta 1$ and $\alpha 6\beta 1$ integrins, but not the collagen receptors $\alpha 1\beta 1$ and $\alpha 2\beta 1$. We have shown recently that an invasin coat replaces the use of Matrigel in 2D organoid sheet cultures⁴⁸. This approach is complementary to the current study: invasin acts fully independently of a hydrogel yet does not work in solution, unlike scTS2/16. Moreover, it replaces the ECM support provided by laminins but not by collagen.

The observation that the activation of $\beta 1$ integrins by scTS2/16 in leukocytes extends to epithelial organoid cells opens a range of applications. The availability of this reagent, which allows organoids to grow in collagen-only hydrogels, may avoid the need for BME or Matrigel. Moreover, its production in *E. coli* or in the HEK293 line allows the generation of clinical-grade batches. Clinical-grade collagen preparations are already widely available for tissue repair³⁷ and cosmetic applications. Mouse and human colon organoids have been shown to be transplantable in mice^{49–51}. To date, two ASC-based early-phase organoid clinical trials have been registered: the transplantation of autologous colon organoids as treatment of ulcerative colitis (clinical trial UMIN000030117, Japanese Registry) and the transplantation of salivary gland organoids for treatment of radiation-induced xerostomia (dry mouth syndrome) (clinical trial [NCT04593589](https://doi.org/10.1038/s41587-025-02874-8)). The use of scTS2/16 would improve expansion of such transplantable human organoids. Moreover, an scTS2/16-induced high-affinity state of the transplanted organoid cells may instruct these to engraft more efficiently.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41587-025-02874-8>.

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Methods

Immunofluorescence analysis of apical–basal polarity and proliferation marker Ki67

P26N human colon organoids were grown for 2 days in BME ± the rat anti-integrin $\beta 1$ antibody A1IB2 (Developmental Studies Hybridoma Bank (DSHB) catalog number A1IB2, Research Resource Identifier (RRID): AB_528306) at a concentration of $1 \mu\text{g ml}^{-1}$. Organoids were collected using cell recovery solution (Thermo Fisher) and fixed in 4% formaldehyde (pH 7.4) at 4°C for 1 h. Hereafter, organoids were washed with PBS 0.1% Tween (PBST) (2% BSA). Three-dimensional imaging of organoids was performed essentially as described previously⁵². In short, organoids were incubated overnight at 4°C with primary antibodies binding ZO-1 (Thermo Fisher Scientific catalog number 61-7300, RRID: AB_2533938), *ITGB1* gene product (R and D Systems catalog number MAB1778, RRID: AB_357909) and Ki67 (Millipore catalog number AB9260, RRID: AB_2142366) or *ITGA2*-encoded gene (Abcam catalog number ab30487, RRID: AB_775702). Following extensive washing with PBST, binding of primary reagents was visualized by overnight incubation in PBST–2% BSA containing donkey anti-rabbit Alexa 568 (Thermo Fisher Scientific catalog number A10042, RRID: AB_2534017) and goat anti-mouse Alexa488 (Thermo Fisher Scientific catalog number A-11070, RRID: AB_2534114). Phalloidin Atto 747N (Sigma-Aldrich) was used for direct visualization of F-actin. DNA was stained with DAPI D1306 (Thermo Fisher). Organoids were optically cleared in a glycerol–fructose clearing solution before imaging. Organoids were imaged on a Leica SP8 confocal microscope. Analysis was done with ImageJ software.

TS2/16 hlgG1 production

We used a simplified workflow to sequence the variable regions of the mouse TS2/16 hybridoma (American Type Culture Collection (ATCC) HB-243) as described⁵³. A human codon-optimized DNA sequence encoding the V_H and V_L domains of antibody TS2/16 was synthesized by GeneArt (Thermo Fisher) and cloned into pFUSEss-CHlg-hG1 and pFUSE2ss-CLlg-hK, respectively (Invivogen). Expression vectors were co-transfected into Expi293F human suspension cells using ExpiFectamine (Thermo Fisher). IgG protein was purified by affinity chromatography using CaptivAHF protein-A affinity resin (Repligen), followed by size-exclusion chromatography using a Superdex 200 Increase 10/300 GL column (Cytiva).

TS2/16 scFv production in *E. coli*

E. coli codon-optimized DNA coding sequences of the TS2/16 scFvs (Supplementary Fig. 3) were synthesized by GeneArt (Thermo Fisher) and cloned into a pET21 vector harboring a C-terminal 6x-HIS tag (Novagen) using standard molecular biology techniques. Protein expression induction, using 1 mM IPTG (Sigma), was done in BL21(DE3) cells starting with an optical density (OD^{600}) of 0.6 in standard lysogeny broth (LB) medium. Production was at 28°C for approximately 18 h. Bacterial pellets obtained were resuspended in lysis buffer (50 mM Tris pH 7.5, 200 mM NaCl, 10% glycerol, 0.4% Triton $\times 100$) and then sonicated (5 cycles; 10 s on, 20 s off). The insoluble fraction, obtained by centrifugation at $20,000 \times g$ for 15 min, was guanidine–hydrochloride solubilized and refolded as described previously²⁰. The partially purified and refolded scFv was further purified using Ni-NTA (Qiagen), followed by size-exclusion chromatography using a Superdex S200 increase 10/300 GL size-exclusion column (Cytiva) in HBS (10 mM HEPES, pH 7.2, 150 mM NaCl). An analysis of the quality of protein purification was performed running samples, under reducing and non-reducing conditions, on a 4–15% TGX precast SDS-PAGE (Biorad) gel. Proteins were visualized using InstantBlue Coomassie Protein stain (Abcam).

scFv TS2/16 production in Expi293 cells

The coding sequence of the TS2/16 scFv (V_H – V_L orientation) containing the artificial human kappa L-chain signal sequence, and C-terminal

tags (8×His tag), HRV 3C site (LEVLFQ/GP), Fc-tag (hulgG1) and a stop codon, was cloned into the pcDNA3.1 mammalian expression vector (Extended Data Fig. 4). Expression was performed by transient transfection of Expi293 cells using the ExpiFectamine transfection kit (Gibco) according to the manufacturer's instruction. After 4 days, the TS2/16 scFv-Fc was captured from the conditioned media with CaptivA HF protein-A affinity resin (Repligen, CA-PRI-0100) and eluted with IgG elution buffer (Thermo Scientific) containing 300 mM NaCl and neutralized with 100 mM Tris pH 8.0 to reach eventually pH 7.0. The sample buffer was exchanged to $1 \times \text{HBS}$ (20 mM HEPES pH 7.3, 300 mM NaCl) using Amicon centrifugal filters (MWCO 30 kDa) and treated with PreScission protease (Cytiva) overnight at 4°C . Next, the Fc-tag was captured with CaptivA HF protein-A affinity resin and the PreScission protease with Glutathione Sepharose 4C (Cytiva). Finally, the TS2/16 scFv was purified on size-exclusion chromatography with Superdex 75 Increase 10/300 GL column (Cytiva) on the ÄKTA pure chromatography system (Cytiva) in $1 \times \text{HBS}$ running buffer (20 mM HEPES pH 7.3, 300 mM NaCl). The TS2/16 scFv was then concentrated using Amicon centrifugal filters (Millipore, MWCO 10 kDa) and sterilized by filtration over a $0.22\text{-}\mu\text{m}$ centrifugal filter (Merck). Protein concentration was determined via NanoDrop A280 reading using the respective protein molecular weight and extinction coefficient. Size-exclusion chromatography and SDS-PAGE analysis for final quality control are shown in (Extended Data Fig. 4).

Integrin $\beta 1$ ectodomain production and immunoprecipitation

We used the procedure reported previously²⁰ to recombinantly produce the extracellular domain of $\beta 1$ subunit in the absence of any α -subunit. Deletion of the ~ 30 amino acid specific-determining loop in the I-like domain is essential for stable secretion of this ectodomain. The DNA coding sequence was synthesized by GeneArt (Thermo Fisher) and cloned into vector pcDNA3.1 (Addgene). Expression vector was transfected into Expi293F human suspension cells using ExpiFectamine (Thermo Fisher). Purification of the C-terminally Flag epitope tagged was performed by absorption of secreted protein to M2-agarose beads (Sigma), and elution by excess $3 \times \text{Flag}$ peptides (Sigma). Eluted proteins were subsequently subjected to size-exclusion chromatography using a Superdex 200 Increase 10/300 GL column (Cytiva). For immunoprecipitation experiments of the $\beta 1$ ectodomain, we used ProteinG (Sigma) or M2-agarose (Sigma). All washes and antibody incubations were in standard PBS. The anti-integrin $\beta 4$ antibody (Abcam catalog number ab110167, RRID: AB_10866385), the anti- $\beta 1$ antibody (R and D Systems catalog number MAB1778, RRID: AB_357909) and the anti-HUTS4 (Millipore catalog number MAB2079Z, RRID: AB_11214024), M2 anti-Flag (Sigma-Aldrich catalog number F3165, RRID: AB_259529), anti-Flag HRP (Abcam catalog number ab49763, RRID: AB_869428) and anti- $6 \times \text{HIS}$ HRP (R and D Systems catalog number MAB050H, RRID: AB_357354) antibodies were used. We used 4–15% SDS-PAGE gel (Biorad) under reducing conditions, and PVDF membrane (Immobilon) for blotting. Enhanced chemiluminescence (ECL) (GE HealthCare) was used in the study.

Adhesion assays of K562 cells

Erythroleukemia cell line K562 cells (Thermo Fisher Scientific) were cultured in DMEM (Gibco) with 100 U ml^{-1} penicillin–streptomycin and 5% FBS (Sigma-Aldrich). Bovine fibronectin (Thermo Fisher) was diluted to $20 \mu\text{g ml}^{-1}$ in PBS and incubated overnight at 4°C in a flat-bottom 96-well plate (Greiner). Subsequently, wells were blocked with 2% BSA in PBS for 30 min at room temperature. K562 cells were resuspended in DMEM/F12 advanced medium (Gibco). Wells were loaded with equal amounts of cells, in the absence of antibodies, or presence of 5 nM of TS2/16 in scFv or IgG format. Following incubation of cells for 90 min at 37°C , unbound cells were removed by three washes with DMEM-F12 advanced medium. The number of adherent cells was analyzed by bright-field microscopic images and a luciferase-based ATP assay (CellTiter-Glo Promega).

FACS staining integrins

Colon (P26N) and duodenum (Duo-72) organoids were enzymatically processed to single cells and incubated with integrin component-specific antibodies, each directly conjugated to a different fluorophore. Stained cells were forward scatter (FSC) and side scatter (SSC) gated to remove debris and to focus the integrin analysis on live cells. Analysis of antibody staining was performed on 10⁴ cells per analysis and compared with the same unstained cells. Stainings were performed by using combinations of the directly fluorochrome-conjugated antibodies listed below:

- (1) Catalog number ab 303014, Biolegend, RRID: AB_493580, APC/cyanine 7 anti-human CD29 antibody Ts2/16 integrin β 1
- (2) Catalog number ab 313619, Biolegend, RRID: AB_2128022, Pacific blue anti-human/mouse CD49f antibody GoH3 integrin α 6
- (3) Catalog number ab 328009, Biolegend, RRID: AB_893368, PE anti-human CD49e antibody SAM-1 integrin α 5
- (4) Catalog number ab 343808, Biolegend, RRID: AB_10641282, APC anti-human CD49c antibody ASC-1 integrin α 3
- (5) Catalog number ab 328311, Biolegend, RRID: AB_2566271, PE/cyanine 7 anti-human CD49a antibody integrin α 1
- (6) Catalog number ab 30486, Abcam, RRID: AB_775704, FITC anti-integrin α 2 antibody (AK7)

Mn²⁺ growth assay

Ileum N39 (ref. 42) organoids cultured in BME were released from the matrix by incubation with dispase (10 U ml⁻¹) at 37 °C (ref. 5). The organoids were then washed with cold DMEM/F12 medium and centrifuged (5 min, 4 °C, 300 × g) to remove residual dispase. The pelleted organoids were treated with TrypLE (no rho kinase inhibitor) at 37 °C to generate cell clusters. This enzymatic reaction was stopped by adding cold DMEM/F12, followed by centrifugation to pellet the cell clusters. The resulting cell clusters were resuspended in 2 mg ml⁻¹ PureCol EZ Gel or BME and plated as small collagen droplets (10–20 μ l). The plate was incubated at 37 °C for 1 h to allow for complete collagen polymerization. After polymerization, the collagen droplets were overlaid with complete expansion medium supplemented with either 1 mM Mn²⁺ or 5 nM scTS2/16. On the first day of the cultures, 10 μ M of rho kinase inhibitor Y-27632 was supplemented to the culture media as well. After day 1, the ileum organoids were maintained in expansion medium supplemented with either 1 mM Mn²⁺ or 5 nM scTS2/16, with the medium refreshed every 2–3 days. On day 7, organoid cultures were imaged using bright-field microscopy followed by quantification of viable organoids using CellTiter-Glo.

N39 ileum FUCCI reporter organoid line

The FUCCI sequence comprising mCherry-Cdt1-T2A-Geminin-hmAzami-Green was PCR amplified from a construct provided as a gift by G. Kops (Hubrecht Institute). With the use of a Gibson Assembly (NEBuilder HiFi DNA Assembly), it was cloned into a p2Tol-based vector harboring puromycin resistance (Addgene category catalog number 714859). Reporter organoids were generated by co-electroporation of 5 μ g of the FUCCI expression construct together with 5 μ g mT2TP transposase mediating the tol2-dependent random integration into the cell's genome. Stable transfectants were isolated based on puromycin resistance and expression of both FUCCI fluorescent signals.

qPCR for intestinal differentiation markers

RNA was extracted using a QIAwave RNA Minikit (catalog number 74534) according to the manufacturer's protocol. cDNA was synthesized using the Applied Biosystems High-Capacity cDNA Reverse Transcription Kit. Real-time PCR was performed on the Biorad CFX96 Real Time PCR System using iQ SYBR Green Supermix 1708880.

The primer sets used were the following:

GAPDH	FWD	GAPDH_FWD	AATGAAGGGGTCATGTATGG
GAPDH	REV	GAPDH_RV	AAGGTGAAGGTCGGAGTCAA
B-actin	FWD	B-actin_FWD	CATGTACGTTGCTATCCAGGC
B-actin	REV	B-actin_REV	CTCCTTAATGTCACGCACGAT
ANPEP	FWD	ANPEP_FWD	GCTGTTTGACGCCATCTCCTAC
ANPEP	REV	ANPEP_RV	GTTCTGGTAGGCAAAGGTGTGG
CYP2C9	FWD	CYP2C9_FWD	CAGAGACGACAAGCACACCCT
CYP2C9	REV	CYP2C9_RV	ATGTGGCTCTGTCTTGCAATGC
CYP2C19	FWD	CYP2C19_FWD	GATCAAAATGGAGAAGGAAAGCA
CYP2C19	REV	CYP2C19_RV	TCAGCTGCAGTGATTACCAAGT
TFF3	FWD	TFF3_FWD	TCCAGCTCTGCTGAGGAGTACG
TFF3	REV	TFF3_REV	ATCCTGGAGTCAAAGCAGCAGC

ITGA2 CRISPR–Cas9 base editing

For single-guide RNA (sgRNA) plasmid preparation, the following sequence was ordered containing a universal priming part and the sgRNA (underlined): TTGTGTAATACTGATCCCCGGTGTTCGTC-CTTCCACAAG. The additional bold cytosine residue is included to obtain sufficient transcription from the U6 promoter. Side-directed mutagenesis in SpCas9-vector 47511 was done using Q5 high-fidelity polymerase with the mentioned primer and the following universal 5'/phos/- GTTTTAGAGCTAGAAATAGCAAGTTAAATAAGGC primer. Following PCR cleanup, T4-ligase and DpnI were used to relegate and generate the plasmid; this was done according to the manufacturer's protocol (NEB). DNA was transformed into DH5 α bacterial cells, and plasmids were verified using Sanger sequencing (Macrogen Europe BV) with sequence primer GGGCAGGAAGAGGGCCTAT.

Organoid electroporation

Organoid electroporation was performed as described previously⁵⁴ with some modifications. Wild-type organoids were maintained in expansion medium. Two days before electroporation, 1.25% (vol/vol) DMSO was added to the organoid medium. Electroporation was done with 10⁶ cells per electroporation with BTXpress solution containing 2.5 μ g *ITGA2* gene directed sgRNA plasmid, 5 μ g plasmid DNA encoding Piggybac transposase, 5 μ g plasmid DNA encoding Piggybac hygromycin cassette⁵⁵ and 7.5 μ g C>T base editor_eGFP plasmid (Addgene plasmid number 112100). Hygromycin 100 μ g μ l⁻¹ (Invivogen) selection was started 5 days post-transfection. Then, 14 days after selection, surviving hygromycin selection clones were individually picked and RT-PCR obtained DNA, sequenced with the following primers: *ITGA2*_fwd GCACCTGCCACCTTCTCCATCA; *ITGA2*_rev GAAAAGAG GAAAGAAGCAGATT. DNA was obtained using a ZYMO Research DNA isolation kit. During clonal outgrowth and selection, scTS2/16 was always present in the media.

Western blot

Proteins were run on a 4–15% SDS-PAGE gel (Biorad) under reducing and non-reducing conditions. Marker proteins were obtained from Thermo Scientific (26616, Pager ruler). SDS-PAGE gel was transferred to a PVDF membrane (Thermo Scientific reference 88518). Transfer efficiency was controlled with Ponceau staining. For blocking, 5% non-fat milk protein in PBS was used. For direct HIS-tagged protein staining, a mouse anti-HIS tag HRP conjugated antibody (R and D Systems catalog number MAB050H, RRID: AB_357354) was used. Indirect staining for the HIS-tag was performed using anti-HIS (Abcam catalog number S2917, RRID: AB_10641949) in combination with anti-mouse-HRP (R and D Systems catalog number MAB050H, RRID: AB_357354). FLAG-tagged proteins were directly visualized using HRP-conjugated M2 anti-FLAG (Abcam, RRID: AB_869428) or indirectly by combining M2

anti-flag (Sigma-Aldrich catalog number F3165, RRID: AB_259529) and anti-mouse-HRP (Abcam catalog number ab6808, RRID: AB_955441). Chemiluminescent substrate ECL (GE-Healthcare) was used.

Bulk RNA sequencing

RNA was extracted using a QIAwave RNA Minikit (catalog number 74534) according to the manufacturer's protocol. RNA integrity was measured using the Agilent RNA 6000 Nano kit with the Agilent 2100 Bioanalyzer, and RNA concentrations were determined using the Qubit RNA HS Assay Kit (Thermo Fisher Scientific). RNA integrity number values of RNA samples were typically 9.5–10.0 and never <9.0. RNA libraries were prepared with the TruSeq Stranded messenger RNA polyA kit and single-end (1 × 50 base pairs) sequenced on an Illumina NextSeq 2000. Reads were mapped to the human GRCh37 genome assembly. Differential gene expression analysis was performed using the DESeq2 package in RStudio (v.2022.02.2). Data visualization was performed using the packages ggplot2 and pheatmap in RStudio or manually plotted using GraphPad Prism (v.8.2.0). The coefficient of determination (R^2) was determined using either the ggplot2 package in RStudio or GraphPad Prism.

BME and PureCol EZ Gel-based 3D organoid culturing

Organoids grown in BME (Cultrex) were passaged by releasing them first from the matrix by incubation for 20 min on ice in cell recovery solution (Thermo Fisher) supplemented with 10 μ M of the Rho-kinase inhibitor Y-27632 (Abmole)⁵. Subsequent DMEM/F12-washed organoids were either mechanically sheared or TrypLE-digested to single cells. Following additional washing with DMEM/F12, organoids were resuspended either in fresh BME or, alternatively, in PureCol EZ Gel (Advanced Biomatrix). For complete gelation of the BME or PureCol EZ Gel gel, plates were incubated at 37 °C for 30 min or 60 min, respectively. Hydrogels were covered with relevant organoid medium and incubated at 37 °C and 5% CO₂. Passaging of organoids growing in collagen was achieved by incubation at 37 °C for 30 min maximally, with collagenase from *Clostridium histolyticum* (Sigma-Aldrich C0130) diluted in DMEM/F12 to a concentration of 1 mg ml⁻¹. For complete removal of the proteolytic enzyme, organoids were washed twice with an excessive volume of DMEM/F12 medium before re-entry into PureCol EZ Gel. The β 1-specific inhibitory A1B2 antibody was added to organoid cultures at 30 nM. The ITGA6-inhibitory antibody G0H3 was added to organoid cultures as 1:1,000 diluted ascites. The Fab versions of SNAKA51, 9EG7 and 12G10 antibodies were provided by J. Li, Harvard Medical School, Boston Children's Hospital. Culture images were taken with an EVOS cell imaging microscope.

BME- and PureCol EZ Gel-based 2D organoid culturing

Transwell inserts (Thincerts, 12-well, pore size 0.4 μ m, Greiner Bio-One 665641) were coated overnight with 2% dilutions of BME (ref. 5) or PureCol EZ Gel in DMEM-F12 medium at 4 °C. Hereafter, inserts were washed twice with DMEM-F12 before the cells were added. Inserts were loaded with approximately $\pm$ 50,000 cells, prepared as a TrypLE-dissociated single-cell suspension. Following digestion, cells were 40- μ m-filtered and washed twice in DMEM-F12 medium in the continuous presence of rho-kinase inhibitor Y-27632 (Abmole). Cells were resuspended in 250 μ l relevant culture medium supplemented with rho-kinase inhibitor Y-27632 (overnight) and optionally long term with 5 nM scTS2/16. Subsequently, 1 ml of identical medium was added to the lower compartment. Passaging of the cultures was started by removing media from both compartments, briefly washing with DMEM-F12 and then filling both compartments with TrypLE.

Organoid differentiation media

Patterning medium⁴² is intestinal medium lacking the p38 inhibitor nicotinamide and PGE2, but has 3 μ M CHIR 99021 (Stemgent 04-0004-base) and 5 nM scTS2/16.

Maturation medium⁴² is patterning medium lacking CHIR 99021 but having recombinant human IL22 (200-22 Peprtech) at a concentration of 2 ng ml⁻¹. Wnt-Surrogate was present at a 30× lower final concentration ($\pm$ 16 pM).

Organoid differentiation in 2D

For immunofluorescence staining, to allow differentiation of organoids, the intestinal expansion medium in the bottom compartment was replaced for 14 days by 'patterning' medium and successively by 'maturation' medium⁴². The upper compartment was exposed to air during this period. Hereafter, the medium was removed, tissue 4% PFA-fixed for 20 min, washed twice with excess PBS and permeabilized by incubating in PBS–0.1% Triton X-100–2% BSA for 1 h. Finally, tissue was washed with PBS 2% BSA. For visualization of enterocytes, we stained for the marker ApoA1 by using rabbit antibody PA5-88109 (Thermo Fisher) (RRID: AB_2804657). Goblet cells were visualized by mouse anti-muc2 antibody 11197 (Abcam) (RRID: AB_297837); for enteroendocrine cells, rabbit antibody 23342-1-AP (Proteintech) detecting chromogranin A (RRID: AB_2879259); and for lysozyme secreted by Paneth cells, rabbit antibody GTX72913 (Gene Tex) (RRID: AB_374812). Alexa-conjugated secondary antibodies (1:500) used were donkey anti-mouse Alexa488 (1:500) Thermo Fisher A21202 (RRID: AB_141607) and donkey anti-rabbit Alexa 568 Thermo Fisher A10042 (RRID: AB_2534017). Actin was stained using Phalloidin-Atto 647 from Sigma. DNA was stained with DAPI D1306 (Thermo Fisher). All reagents were diluted in PBS–2% BSA.

PureCol EZ Gel polymerization conditions

Ileum N39 (ref. 42) organoids cultured in BME were first released from the matrix protein using dispase (10 U ml⁻¹)³⁸. The organoids were incubated with dispase at 37 °C for 30 min, after which they were washed twice with cold DMEM/F12 and collected by centrifugation (5 min, 4 °C, 300 × g). The washed organoids were then treated with TrypLE (without rho kinase inhibitor) at 37 °C for 5 min to dissociate the organoids into clusters of cells. To stop enzymatic activity, the cell suspension was washed again with cold DMEM/F12 and centrifuged (5 min, 4 °C, 300 g). The resulting cell pellet was resuspended in 1 ml of one of the following conditions: DMEM/F12, complete expansion medium (EM), EM + Rho kinase inhibitor Y-27632 or EM + Rho kinase inhibitor Y-27632 + scTS2/16. The cell suspension was herein incubated for 30 min (room temperature), allowing conditioning before embedding in hydrogel. After this incubation, cells were centrifuged (5 min, 4 °C, 300 × g) and cell pellets resuspended in 2 mg ml⁻¹ collagen EZcol hydrogel. The suspension was plated in 10–20- μ l domes which were then incubated at 37 °C for 1 h to allow polymerization. Hereafter, cultures were covered with EM supplemented with rho kinase inhibitor Y-27632 and 5 nM scTS2/16. The next day, Y-27632 was removed from the medium, and cultures were maintained in EM with 5 nM scTS2/16 to analyze the effect of diffusion of scFv in collagen hydrogels. After 5 days of culture, the number of outgrown organoids was quantified using CellTiter-Glo viability assay (Promega).

Long-term 3D PureCol EZ Gel (Fujii medium)

Ileum (N39) organoids were cultured in expansion medium consisting of DMEM/F12 supplemented with 100 U ml⁻¹ P/S, 10 mM HEPES, 1× Glutamax, 2% B-27 with Vita, 3% (v/v) recombinant R-spondin-3, 2% (v/v) recombinant noggin, 1 mM N-acetylcysteine, 0.5 nM Wnt surrogate, 0.1 μ g ml⁻¹ hGF-1 (Peprtech), 0.05 μ g ml⁻¹ hGF2 (Peprtech), 10 nM gastrin, 0.5 μ M A83-01, 0.05 μ g ml⁻¹ hEGF, 3 μ M CHIR 99021 and 10 μ M forskolin (Tocris)⁴². For collagen cultures, ileum organoids grown in BME were first released from the matrix by incubation with dispase (10 U ml⁻¹ for 30 min at 37 °C). Hereafter, organoids were washed twice with cold DMEM/F12 and collected by centrifugation at 300 × g for 5 min at 4 °C. Organoids were then treated with TrypLE (no rho kinase inhibitor) at 37 °C for 5 min and further disrupted mechanically to

dissociate organoids into small clumps of cells (avoiding complete single-cell dissociation). Organoids were washed again with cold DMEM/F12, centrifuged under the same conditions and resuspended in 2 mg ml⁻¹ PureCol EZ Gel. The matrix–cell suspension was seeded in 10–20-μl droplets onto a culture plate, which was subsequently incubated at 37 °C for 1 h to allow polymerization. Hereafter, the droplets were covered with the expansion medium supplemented with 10 μM Y-27632 rho-kinase inhibitor (initial 12 h) and 5 nM scTS2/16. The following day, this medium was refreshed, with the expansion medium lacking rho kinase inhibitor but continuing with 5 nM scTS2/16. This medium was refreshed every 2–3 days. For passaging of organoids, the collagen domes were incubated with 1 mg ml⁻¹ collagenase from *Clostridium histolyticum* (Sigma-Aldrich) for 30 min at 37 °C, and digestion monitored using bright-field microscopy (small clumps of cells). Organoids were collected in a 15-ml tube and washed twice (centrifugation for 5 min at 300 × g) with an excess volume of cold DMEM/F12. The clumps of cells were washed twice and resuspended in fresh PureCol EZ Gel (2 mg ml⁻¹).

3D collagen differentiation protocol

The ileum N39 triple reporter line, with fluorescently marked specific intestinal cell types (goblet cells: Muc2-GFP; enteroendocrine (EEC) cells: CHGA-iRFP; and Paneth cells: DEFA5-dsRED), was described⁴². It was propagated in 3D hydrogel. At both passage 0 (P0) and passage 2 (P2), the organoids were induced to differentiate. Then, 7 days following passaging, the expansion medium was replaced with differentiation medium in all 3 hydrogel conditions: (1) collagen, (2) collagen + scTS2/16 and (3) BME. The differentiation (patterning) medium was prepared as described previously⁴². The differentiation process was monitored daily using EVOS fluorescence microscopy to track the emergence of the differently marked mature epithelial types. After 5 days, mature epithelial cells were observed in the collagen- and BME-cultured organoids. At this point, organoids in all hydrogel conditions were fixed to maintain consistency in imaging and downstream analyses. For quantitative assessment of differentiated secretory cells, we used flow cytometry (BD Fortessa). A total of 5,000 DAPI-negative single cells per hydrogel condition were analyzed to compare differentiation efficiencies to the expansion culture. Organoids were fixed in 4% formaldehyde (pH 7.4) and subsequently analyzed using SP8 confocal microscopy.

Organoid lines

- The human small-intestinal duodenum line Duo-72 was described⁵⁶.
- Jejunum J2 organoids were provided by M. Estes, Baylor College of Medicine. Organoids were described⁵⁷.
- Ileum N39 organoids were described⁵⁸.
- The ileum N39 triple reporter was described⁴².

- Primary colon biopsies (Fig. 5e, N = 4) were derived from the Utrecht Portal for Organoid Technology. Patients (University Medical Center Utrecht (UMCU)) provided informed consent for the generation of such organoids based on protocol UPORT Cancer Biobank 21-042 and was in accordance with the Declaration of Helsinki and Dutch law. This study complies with all relevant ethical regulations regarding research involving human participants.
- P26N colon organoids were described⁴⁷.
- Primary liver human ductal organoids (Fig. 5f, N = 3) were derived from UMC biopsy codes 057932, 061530 and 061500. The study was approved by the UMCU ethical committee (TCBio 21-042) and was in accordance with the Declaration of Helsinki and Dutch law. This study complies with all relevant ethical regulations regarding research involving human participants.
- The ductal pancreas line, code HUB-08-A2-007, was obtained from the biobank of UMCU (Toetsingscommissie Biobanken TCBio (Biobank research Ethic Committee)) approved by protocol 12-093 HUB-Cancer and informed donor consent.
- Established human liver and pancreas ductal lines were described earlier⁵⁹.

All patients participating in this study signed informed consent forms approved by the responsible authority. In all cases, patients can withdraw their consent at any time, leading to the prompt disposal of their tissue and any derived material. Future distribution of organoids to an academic party will have to be authorized by the Medical Ethics Committee of the UMCU (TCBio).

Statistics

All effects measured between two groups, in response to agonistic (TS2/16) or antagonistic (AIIB2 or G0H3) antibodies, such as CellTiterGLO units—number of cells—or fold-change differences in gene expression levels in qPCR, were tested for significance using one-tailed *t*-tests. *P* values and the number and type of replicates (designated *n*) are provided in the relevant legends of the main figures, extended data figures and supplementary figures. Significance of results from bulk sequence comparisons (Extended Data Figs. 9 and 10) were analyzed using a Wald test for negative binomial distribution in the expression between scFv and control samples. Bonferroni correction was applied to adjust for multiple testing. The number of independent identical experiments is designated *N* in all legends.

Organoid expansion medium

Organoid expansion medium for each organoid type included the following components: a ‘+’ indicates that the component is present in the medium for the corresponding organoid.

Reagent	Source	Concentration	Intestinal	Stomach	Liver ductal	Pancreas ductal
F12 advanced	Gibco	Not applicable	+	+	+	+
HEPES	Gibco	10 nM	+	+	+	+
Glutamax	Gibco	1:100	+	+	+	+
N-acetylcysteine	Sigma-Aldrich	1.25 mM	+	+	+	+
Nicotinamide	Sigma-Aldrich	10 μM	+		+	+
B27	Sigma-Aldrich	1:50	+	+		+
p38-inhibitor SB202190	Sigma-Aldrich	10 μM	+			
A83-01 inhibitor TGFβ type I receptor	Tocris	0.5 μM	+	+	+	+
PGE2	Tocris	1 μM	+			+
Gastrin	Tocris	10 nM		+	+	+
Forskolin	Tocris	10 μM			+	
EGF	Peptrotech	50 ng ml ⁻¹	+	+		+